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MIET2499 – Stress Analysis

BRIDGE DESIGN REPORT

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Due date: 25/09/2024

September 2024

School of Science, Engineering, and Technology

What's next...

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1. Introduction & Theory

a) In this project, our team is necessitated to conceptualize, draft, and complete the design for constructing a Acrylic Plastic bridge that meets the specified criteria set by the lecturer. Our team, is basically called team 5, is tasked with achieving this by utilizing the knowledge acquired from Week 1 to 9 of the Stress Analysis course. The objective of this project is to enhance our understanding of calculating bending stress and moment, shear stress, and the deflection of a structure under various load conditions.

The lecturer's **specified requirements** are as follows:

- The combined mass of two PAScars and iron blocks is 8.5kg, with a safety factor of 1.5.
- The vehicle, carrying the blocks, must traverse an unsupported span of 900mm on the bridge. Guard rails are required to prevent the vehicle from falling off.
- Two PASCO dynamic tracks, each 1.2 meters in length, are positioned on either side of the table. The bridge must ensure a smooth transition for the moving masses between the bridge and the dynamic tracks.
- The bridge features a central gap to permit a tool attached on a vehicle to access the area beneath the bridge, during the designing of the bridge, this gap must remain unobstructed at all times.

- The cross-sectional area of the bridge, including the guard rails, must be symmetrical.
- The bridge will be fixed at one end by the 2 hand-clamps (first 50mm length) and simple support will be provided by the table surface where the bridge terminates.
- The team name must be embossed on the bridge surface for identification purposes.
- The bridge must remain intact and undamaged during the demonstration.

b) The subsequent parameters are essential to the project, and their magnitude, direction, or location must be ascertained:

- Maximum deflection (v)
- Bending moment stress (σ)
- Shear Stress (τ)
- Maximum bending moment (M_{max})
- Maximum shear force (V_{max})

During this project, we will utilize the following theorems, formulas, and statistical data:

Static equilibrium:

$$\Sigma F_{xy} = 0$$

$$\Sigma M = 0$$

Calculation for allowable stress:

$$\text{Safety factor F.S} = \frac{F_{yield}}{F_{allow}} = \frac{\sigma_{yield}}{\sigma_{allow}} = \frac{\tau_{yield}}{\tau_{allow}}$$

Average normal shear stress:

$$\tau = \frac{V}{A}$$

Sectional modulus:

$$\text{S} = \frac{M_{max}}{\sigma_{allow}} = \frac{I}{C}$$

Unstationary load:

The objects on the bridge are dynamic, moving along the track and changing positions. Consequently, the forces and moments at the supports become variables rather than constants. This introduces a challenge in calculations comparing to most classroom problems that involve stationary loads. The most straightforward approach to address this issue is to introduce a variable "a", which represents the distance from the left support (the starting point of the car) to the

position of the left-pair-wheels. The forces and moments at the supports will then be expressed as functions of this variable "a".

Shear force and Moment Diagram (V-M Diagram):

In the context of dead loads, shear force and moment diagrams effectively indicate the locations of maximum shear force and moment, as the support forces and moments remain constant. However, for a moving load with four pairs of wheels, each position of "a" results in distinct shear force and moment diagrams. To faciliate this process, the DESMOS web application is employed to graph the functions and pinpoint the exact positions of the objects where V_{max} and M_{max} are identified.

Moment of Inertia I:

Generally looking at our design, the bridge is not a completely rectangle. As a result, we will employ the formula $I = I + Ad^{2,}$ where $I = \frac{bh^3}{12}$, A =bh, and "d" represents the distance between the centroid y bar and centroid smaller sub-areas. The strategy is sectioning the cross-sectional area into smaller areas , calculate the moment of inertia I_i for each small area, and then aggregate these values to obtain the total I.

Flexural Rigidity:

Flexural rigidity, denoted as EI, is a product if Young's Modulus E and the moment of inertia I, which needs to be determined. A higher value of EI enhances the structure's resistance to deflection.

Bending moment stress:

The maximum stress, σ_{max}, is given by

$$\sigma_{max} = \frac{M * y}{I}$$

Where "M" represents the maximum moment, "y" is the distance from the centroid to the extreme point of the bridge, and "I" is the moment of inertia. An increase in "y" results in a higher stress.

Formulas for calculating the slopes and deflections of beams using the superposition method:

Material (Acrylic Plastic) given parameters:

- Young's modulus: 1.1 Gpa
- Yield strength: 24.4 Mpa
- Shear yield strength: 14.1 Mpa
- Density: 0.043 (lb/in^3) – 1.18 (g/cm^3)
- Poisson's ratio: 0.37

The approach to solve this bridge problem involves the following steps:

1. Determine the allowable bending moment and shear stress by dividing the given shear yield and bending stress by the safety factor of 1.5.
2. Determine the moment of inertia (I).
3. Calculate the reaction forces and moment Ay, By, Ma, where A is the fixed support end and B is the roller support end of the bridge, based on the variable "a". Here, "a" represents the distance from the fixed support (starting point of the cars) to the left-pair wheels, using the superposition method.
4. Apply the superposition method and the formulas for deflection at the endpoint, distributed load, and intermediate load to achieve the maximum deflection. If the deflection is less than 7mm, the stiffness condition is satisfied.
5. Utilize the section method to derive formulas for shear force and moments in each section of the bridge. Input these equations into DEMOS and adjust "a" until the maximum shear force and moment are indentified.
6. Calculate the shear and bending stress, then compare them to the allowable stresses. If the calculated stresses are lower than the allowable stresses, the design is deemed acceptable.

c) During our project, we have encountered several potential erros that can be categorized as following:

- **Measurement error:** This error is typically arised from one individual's perspective during measurement and the quality of the measurement tool. For instance, during the measurement for the dynamic track to design the height for the bridge, one member used a standard ruler and recorded a height of 2.5 cm. However, another member used a caliper and obtained a more precise measurement of 2.3 cm.

- **Assembly and Tolerance error:** After receiving the parts from the workshop, we started to glueing to attach the parts together. However, the glue occupied more space than expected, resulting in the bridge appearing aesthetically unpleasing.
- **Analyzing error:** This is a type of error that every engineers need to avoid, as it can lead to consistently incorrect results throughout the calculation process. This problem could be seen by the presence of anomalous results during calculations, indicating that something is amiss.

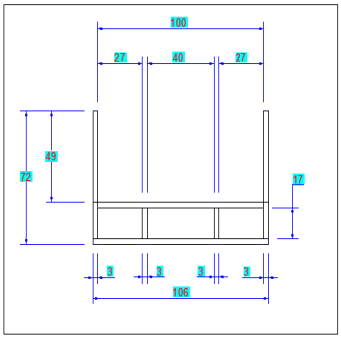
d) The lecturer has outlined several constraints, necessitating an increase in the cross-sectional area to reduce stress, despite limited material availability:

- The bridge must be constructed completely from Acrylic with a thickness of 3mm, and its total surface area must not exceed 2500 cm^2. This restriction limits the cross-sectional area, thereby increasing stress and reducing the moment of inertia. Given these constraints, it is essential to maximize material usage to increase the cross-sectional area while maintaing uniform cross-sectional cuts. This approach makes it easier to coordinate and reduces the amount of calculations needed, which saves both time and resources.
- The bridge must span a gap 900mm, meaning "L" in the deflection formula is 0.9m. A larger "L" results in greater deflection unless intermediate supports are used.
- In the design and assembly process, the central gap in the bridge must remain unobstructed. This introduces few more challenges in critical thinking regarding to calculations and designing.
- On one end of the bridge will be fixed using hand clamps, while the other end will rest on the table to stimulate a roller support. The fixed support at end A introduces a moment, changing the sectional moment wherever we cut along the bridge and thereby decreasing bending stress.
- The maximum deflection of the bridge, as the PAScars traverse it, must not exceed 7mm. Thus, the total deflection calculated using theoretical principles must be less than 0.007m.
- The combined weight of the two PAScars and iron blocks is 8.5kg. This weight influences shear stress, bending stress, and deflection, with greater weight leading to increase stress and deflection. The only viable solution is to increase the cross-sectional area.

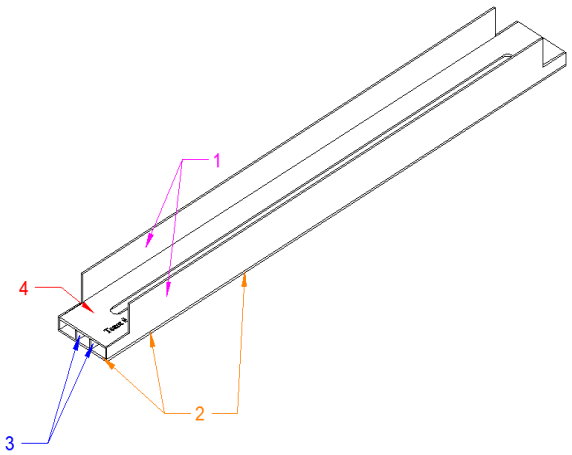
2. Design Proposal

2.1. Conceptual Design

The team's bridge design is expected to utilize a total of 2494.85 cm^2 / 2500 cm^2 of material. From the provided bridge deck the lecturer gave to the team, we will incorporate only three additional components, maintaining as simplest as we can. The primary component is the "under_deck", it has dimension 1000 x 106 x 3 mm, serves as the foundational base for the other components. Next, we design two guard rails on either side prevent the car from deviating off the bridge path. These guard rails not only keep the PAScars on track but also enhance the bridge's moment of inertia. The taller the guard rails, the increasing the cross-sectional area and reducing stress. Addtionally, two vertical support beams were added to reinforce the bridge deck from bending, they will be positioned between the stock piece and the under_deck to augment the area and establish a connection. Silicon glue, will be applied using silicon guns purchased from stationery shops, was chosen to adhere the components together.



ITEM NO.	PART NAME	DESCRIPTION	SURFACE AREA(cm^2)	Q.Ty
1	Rail Bridge	Prevent cars from going out the path on the bridge.	641	2
2	Under_deck	Strengthen the underneath of the bridge	872.85	1
3	Support	Vertical support on the Under_deck part, extra strengthen the bridge	170	2
4	Given_deck	The provided deck	825.43	1



Course coordinator	Dr.Thanh	MIET2499–Stress Analysis
Team	Team 5	
Bridge design	20/9/2024	Unit: mm
		Ratio : 1/1

Figure 2.2: The overall conceptual design

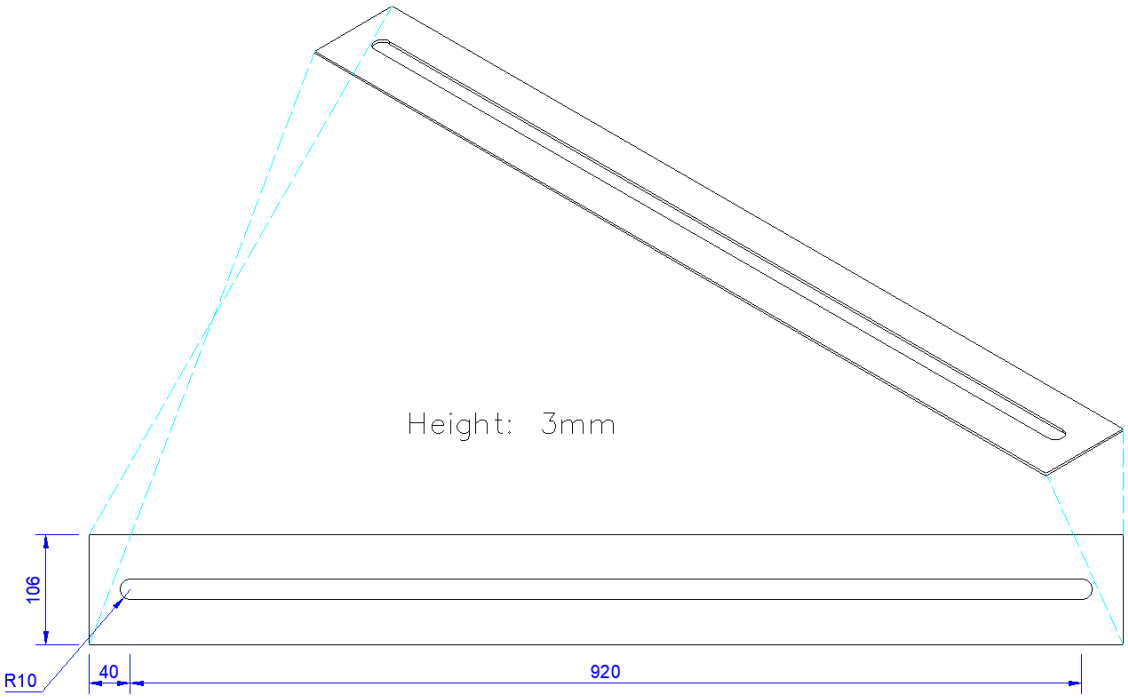


Figure 2.3: Top and Isometric view of the conceptual Under_deck component

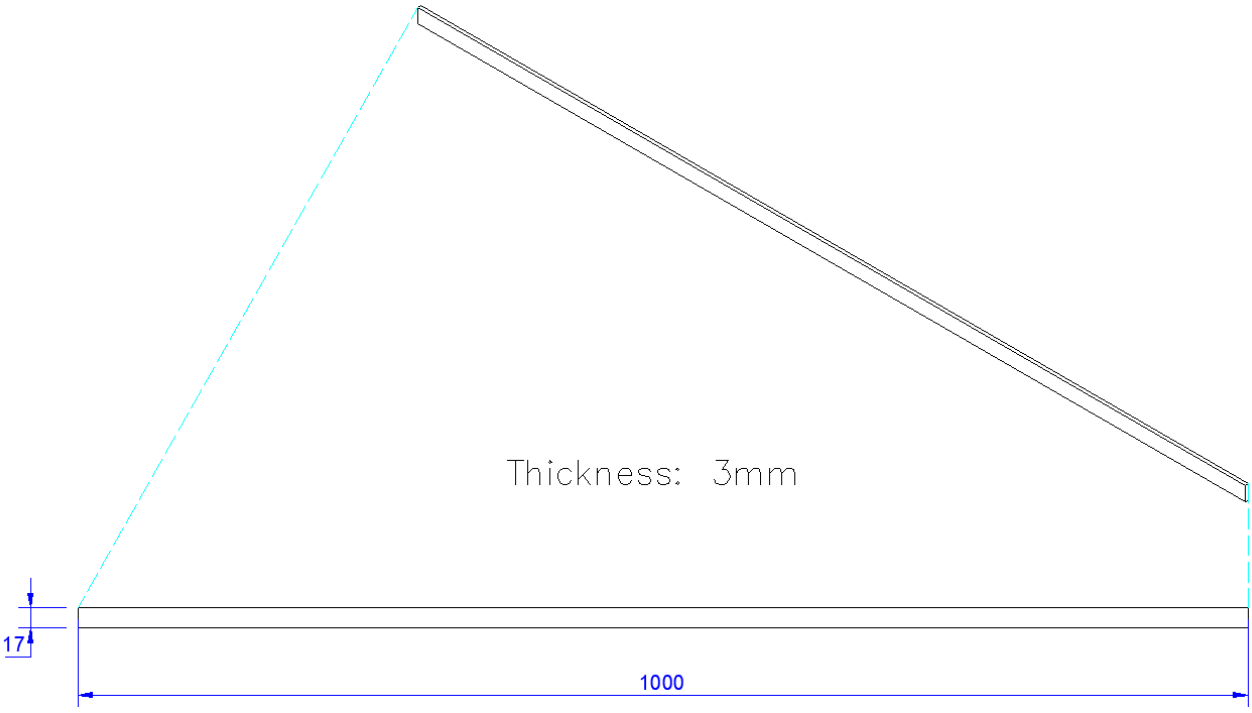


Figure 2.4: Side and Isometric views of the conceptual support beam

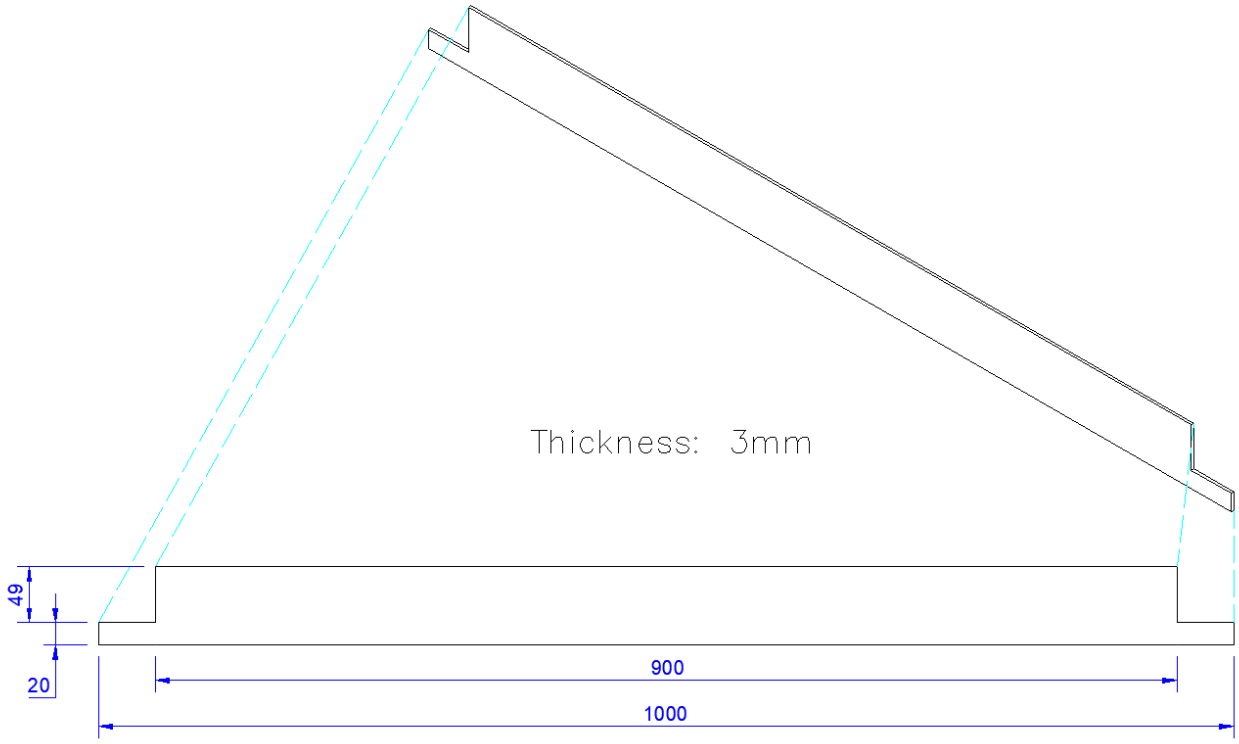


Figure 2.5: Side and Isometric views of the conceptual Rail Bridge

2.2. Materials:

The total material usage for this construction is expected to be 2494.85 cm^2. The bridge parts will be created by laser cutting. In addition, they are made from acrylic plastic (with a thickness of 3mm). We are going to assemble all the parts of the bridge with silicone glue and used a hot melt glue gun



Figure 2.6: Silicon glue gun and tape measure will be applied.



Figure 2.7: Toy trucks and loads be applied for the demonstration's day.

2.3. Technical Draw/Sketch in Details

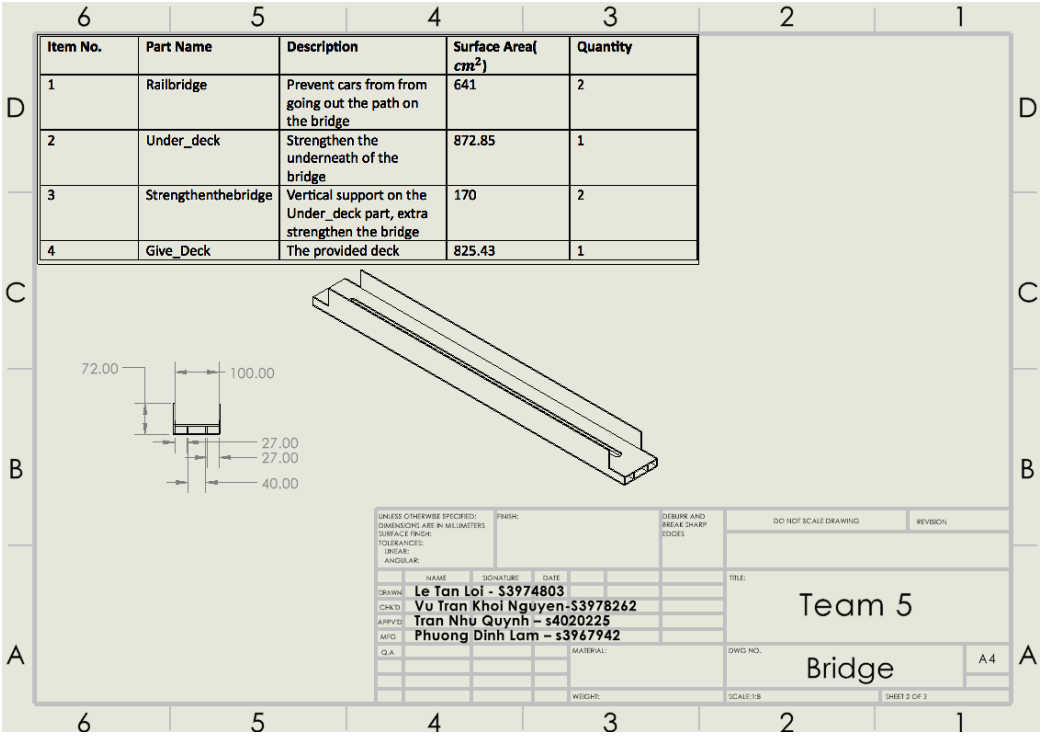


Figure 2.8: Front and Isometric Views of the conceptual bridge conducted by Solidworks drawing.



Figure 2.9: Top and Side Views of the conceptual bridge.

3. Design and Analysis

3.1. Design Load Cases

According to our design, our team members ended with the following load cases that we considered ” worst-case scenarios”:

Load case 1: Maximum bending stress

Load case 2: Maximum shear stress

Load case 3: Maximum deflection

Load case 4: The dynamic load cases caused by the moving 2 PAScars carrying the 8kg load.

Load case 5: The constant load appeared from the weight of the bridge itself.

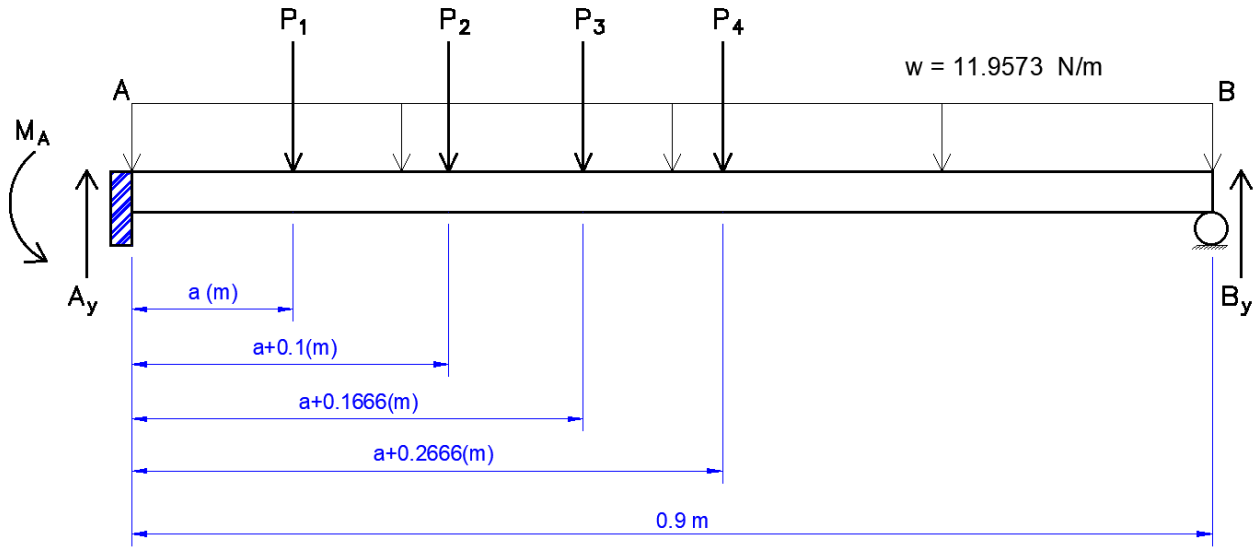


Figure 3.1: Illustration of designing load cases

The reasons we selected such these load cases as the "worst-case scenarios":

Load case 1: This case examines the bridge's ability to withstand bending moments induced by substantial point loads, which may results in structural failure if the material's yield strength is surpassed.

Load case 2: Examining shear stress is essential for comprehending the distribution of forces across a bridge, specifically at the supports. A uniformly distributed load replicates real-world scenarios where various vehicles or environmental factors can exert forces concurrently, thereby maximizing shear forces.

Load case 3: Significant deflection can indicate potential issues in the design and adversely affect the bridge's serviceability. Evaluating the bridge under extreme load conditions is essential to confirm its functionality and safety during peak operational scenarios.

Load case 4: Ensuring the bridge's capacity to accommodate 2 PAScars combined with 8kg load is crucial and guarantees safety during standard operational conditions.

Load case 5: Essential for the fundamental structural support needed for the bridge, this knowledge guides the designing of the supporting components.

3.2. Maximum Stress Analysis

By using superposition, bending moment of the beam can be express as:

- $M = M_w + M_1 + M_2 + M_3 + M_4 + M_{By}$ $for\ (x \leq a)$
- $M = M_w + M_2 + M_3 + M_4 + M_{By}$ $for\ (a \leq x \leq a + 0.1)$
- $M = M_w + M_3 + M_4 + M_{By}$ $for\ (a + 0.1 \leq x \leq a + 0.1666)$
- $M = M_w + M_4 + M_{By}$ $for\ (a + 0.1666 \leq x \leq a + 0.2666)$
- $M = M_w + M_{By}$ $for\ (a + 0.2666 \leq x \leq 0.9)$

By using superposition, shear force of the beam can be express as:

- $V = V_w + V_1 + V_2 + V_3 + V_4 + V_{By}$ $for\ (x \leq a)$
- $V = V_w + V_2 + V_3 + V_4 + V_{By}$ $for\ (a \leq x \leq a + 0.1)$
- $V = V_w + V_3 + V_4 + V_{By}$ $for\ (a + 0.1 \leq x \leq a + 0.1666)$
- $V = V_w + V_4 + V_{By}$ $for\ (a + 0.1666 \leq x \leq a + 0.2666)$
- $V = V_w + V_{By}$ $for\ (a + 0.2666 \leq x \leq 0.9)$

The analysis to find out the bending moment, shear force of the beam is stated in the appendix

Using Desmos, we can graph bending moment and shear force diagram as follow:

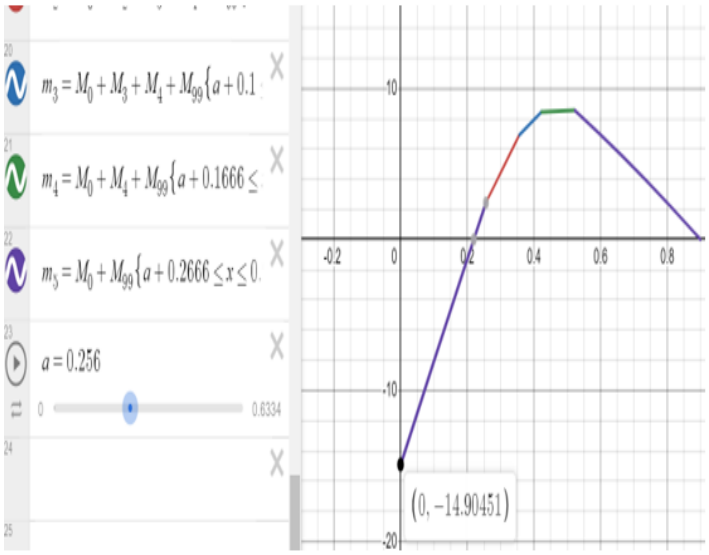


Figure 3.2: Bending moment diagram conducted by DEMOS

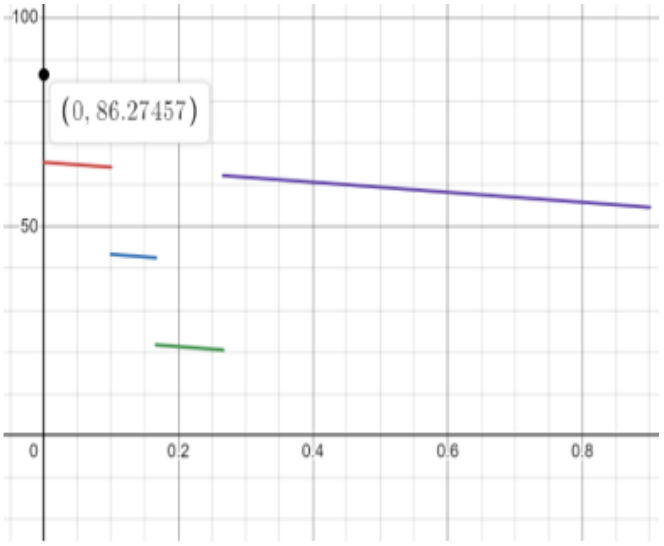


Figure 3.3: Shear force diagram conducted by DEMOS

According to figure 10, we have maximum bending moment at $x = 0\ m;\ M_{max} = 14.905Nm.$

According to figure 11, we have maximum shear stress at $x = 0m$ or $x = 0.9m$ then $V_{max} = 86.275N$

Therefore, maximum shear stress is:

$$\tau_{max} = \frac{V_{max}Q}{It} = \frac{86.275 * 7.564 * 10^{-4}}{3.863 * 10^{-7} * (106 - 20) * 10^{-3}} = 19642.423N$$

$$\tau_{allow} = \frac{\tau}{safety\ factor} = \frac{14.1}{1.5} = 9.4\ MPa > \tau_{max}$$

Therefore, maximum normal force resulted by bending moment is:

- Maximum compression:

$$\sigma_{max} = -\frac{M_{max}y}{I} = -\frac{14.905 * (72 - 21.938) * 10^{-3}}{3.863 * 10^{-7}} = -1931592.312N$$

- Maximum tension:

$$\sigma_{max} = -\frac{M_{max}y}{I} = -\frac{14.905 * (0 - 21.938) * 10^{-3}}{3.863 * 10^{-7}} = 846455.8374N$$

$$\sigma_{allow} = \frac{\sigma}{safety\ factor} = \frac{24.4}{1.5} = 16.27\ MPa > \sigma_{max}$$

All stresses are much smaller than allowable stresses, therefore, the bridge can withstand.

3.3. Deflection Analysis :

Using deflection formulas found in the section above we can easily obtain deflection of the bridge:

$v_1 = v_w + v_{P1,2} + v_{P2,2} + v_{P3,2} + v_{P4,2} + v_{By}\ for\ (x \le a)$

$v_2 = v_w + v_{P1,1} + v_{P2,2} + v_{P3,2} + v_{P4,2} + v_{By}\ for\ (a \le x \le a + 0.1666)$

$v_3 = v_w + v_{P1,1} + v_{P2,1} + v_{P3,2} + v_{P4,2} + v_{By}\ for\ (a + 0.1 \le x \le a + 0.1666)$

$v_4 = v_w + v_{P1,1} + v_{P2,1} + v_{P3,1} + v_{P4,2} + v_{By}\ for\ (a + 0.1666 \le x \le a + 0.2666)$

$v_5 = v_w + v_{P1,1} + v_{P2,1} + v_{P3,1} + v_{P4,1} + v_{By}\ for\ (a + 0.2666 \le x \le 0.9)$

The analysis to find deflection resulted by all loads is stated in the appendix.

With "a" ranging from 0 to 0.9 – 0.2666 = 0.6334 since we only consider case when car fully on the bridge, by substituting By founded in section 3.3 in to deflection equation v_{By} and graph v_1, v_2, v_3, v_4, v_5 using DESMOS (appendix) we can see the largest deflection is: $1.380mm$ at $x = 0.524m$

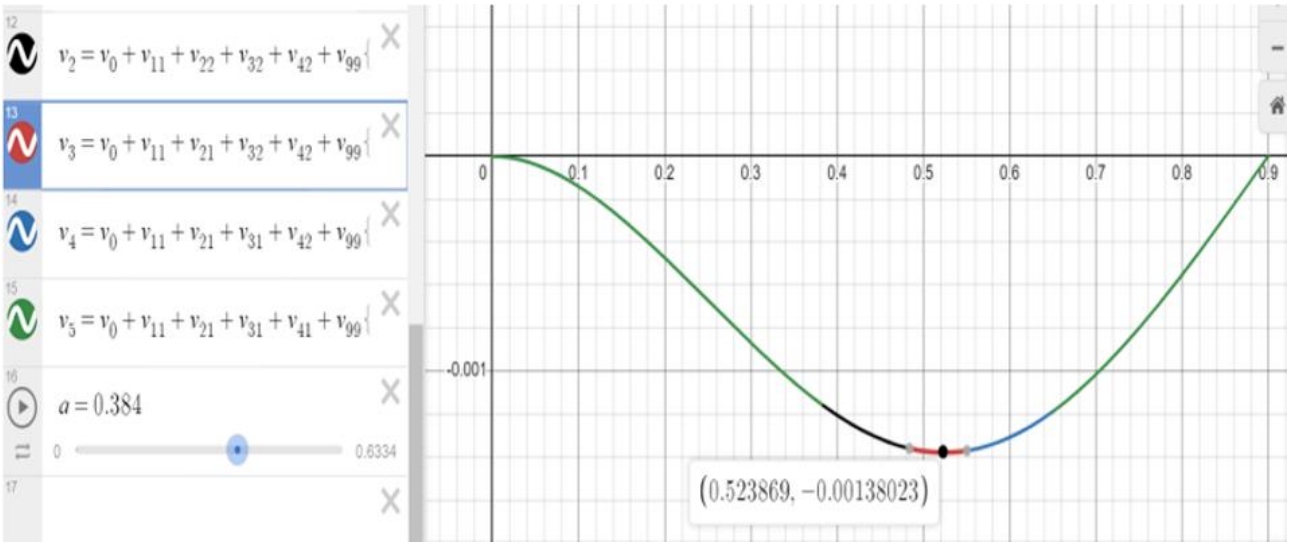


Figure 3.4: Deflection of beam conducted by DEMOS

4. Conclusion and Future works :

4.1. Main outcomes and Discussion

Longitudinal shear stress of the beam can be calculated as: $\tau = \frac{V_{max}Q}{It}$, by substituting $V_{max}.Q$ into the equation, we have : $\tau_{max} = 19642.423N$. Maximum shear stress in the bridge occurs when the PAScars are positioned at the supports, specifically at the beginning or the end of the bridge a = 0m or a = 0.6334m. To ensure the bridge from failing due to the longitudinal shear stress, so it's crucial to design the bridge's cross-sectional area to adequately resist shear forces. Choosing materials with high shear strength and incorporate shear reinforcements like stirrups or

cross-bracing to enhance shear resistance. Of particular note is that acording to the shear stress formula, we increase the cross -sectional area.

Maximum bending stress occurs when the car is at $a = 0.256m$ from the fixed end of the bridge. Then $\sigma_{compression} = -1931592.312N$ and $\sigma_{tension} = 846455.8374N$. They are less than the Allowable Bending Stress = 16.27 MPa , therefore, the bridge can withstand. To prevent failure due to Bending Stress in a bridge, it's essential to design for sufficient strength and stability. Since the formula stress can only be minimized by dividing by a larger I (inertia), we prevent stress failure by increasing I (inertia), the moment of inertia.

Maximum deflections occurs when car at $a = 0.384m$ from fixed end of the bridge. Then the bridge deflected 1.380mm downward. This means that it satisfies the requirement that the maximum deflection should be less than 7mm. To ensure that the bridge does not fail due to deflection or stiffness condition, it's essential to ensure adequate stiffness and limit deflection. So, increased flexural rigidity (EI) is necessary to prevent deflection failures since load deflection formulas only allow for this one element to decrease deflection. Since we are limited to one material and are unable to alter it, E (modulus of elasticity) in our instance is constant. Addition of more materials can increase I (inertia).

Proposal design:

- Following extensive calculation we found out that the moment inertia of the conceptual design satisfies all the requirement outlined in the project description. Additionally, the calculations showed that the deflection of the conceptual design bridge is $1.380mm$ at the position $x = 0.524m$ which is well within the maximum allowable deflection of 7mm in the project description. As a result, our team decided to proceed with the conceptual design on the demonstration day. In this case, the moment of inertia remains :

  $$\mathbf{I = 3.863*10^{-7}\ (m^4)}$$

  The analysis to find out the moment of inertia is stated in the appendix.

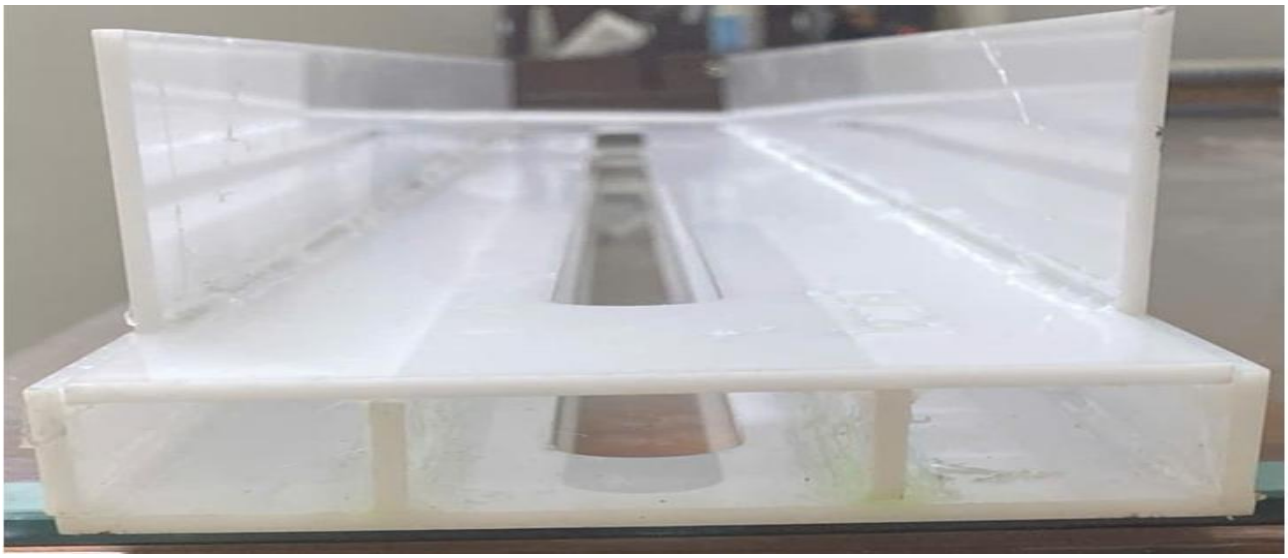


Figure 4.1: The captured actual cross-sectional area of the bridge

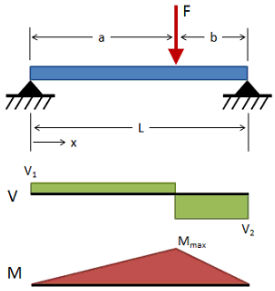


Figure 4.2: The actual deflection of the bridge captured during the Demonstration day

4.2. Comparison to other cases

By applying the same analysis procedure, specifically the superposition method, we can derive the equations for bending moment, shear, and deflection of the bridge. These equations can then be graphed using DESMOS.

Simply Supported, Intermediate Load



Deflection:

$$\delta = -\frac{Fbx}{6LEI}\left(L^2 - b^2 - x^2\right) \qquad (0 \le x \le a)$$

For $a \ge b$:

$$\delta_{max} = \frac{Fb(L^2 - b^2)^{3/2}}{9\sqrt{3}LEI} \qquad @\ x = \sqrt{\frac{L^2 - b^2}{3}}$$

Slope:

$$\theta = -\frac{Fb}{6LEI}\left(L^2 - b^2 - 3x^2\right) \qquad (0 \le x \le a)$$

$$\theta_1 = -\frac{Fab(L+b)}{6LEI} \qquad @\ x = 0$$

$$\theta_2 = \frac{Fab(L+a)}{6LEI} \qquad @\ x = L$$

Shear:

$$V_1 = +Fb/L \qquad (0 \le x \le a)$$

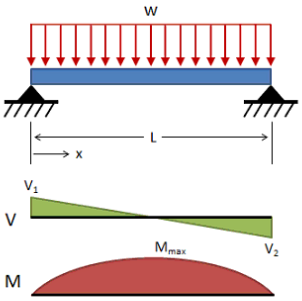
$$V_2 = -Fa/L \qquad (a \le x \le L)$$

Moment:

$$M_{max} = +Fab/L \qquad @\ x = a$$

Figure 4.3: Point load stress and deflection formula

Simply Supported, Uniform Distributed Load



Deflection:

$$\delta = -\frac{wx}{24EI}\left(L^3 - 2Lx^2 + x^3\right)$$

$$\delta_{max} = \frac{5wL^4}{384EI} \qquad @\ x = L/2$$

Slope:

$$\theta = -\frac{w}{24EI}\left(L^3 - 6Lx^2 + 4x^3\right)$$

$$\theta_1 = -\frac{wL^3}{24EI} \qquad @\ x = 0$$

$$\theta_2 = +\frac{wL^3}{24EI} \qquad @\ x = L$$

Shear:

$$V = w\,(L/2 - x)$$

$$V_1 = +wL/2 \qquad @\ x = 0$$

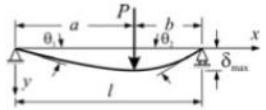
$$V_2 = -wL/2 \qquad @\ x = L$$

Moment:

$$M_{max} = wL^2/8 \qquad @\ x = L/2$$

Figure 4.4: Uniform load stress and deflection formula [6]

7. Beam Simply Supported at Ends – Concentrated load P at any point



$$\theta_1 = \frac{Pb(l^2 - b^2)}{6lEI}$$

$$\theta_2 = \frac{Pab(2l - b)}{6lEI}$$

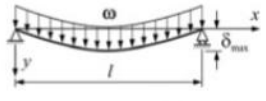
$$y = \frac{Pbx}{6lEI}\left(l^2 - x^2 - b^2\right) \text{ for } 0 < x < a$$

$$y = \frac{Pb}{6lEI}\left[\frac{l}{b}(x-a)^3 + \left(l^2 - b^2\right)x - x^3\right] \text{ for } a < x < l$$

$$\delta_{max} = \frac{Pb\left(l^2 - b^2\right)^{3/2}}{9\sqrt{3}\,lEI} \text{ at } x = \sqrt{\left(l^2 - b^2\right)/3}$$

$$\delta = \frac{Pb}{48EI}\left(3l^2 - 4b^2\right) \text{ at the center, if } a > b$$

8. Beam Simply Supported at Ends – Uniformly distributed load ω (N/m)



$$\theta_1 = \theta_2 = \frac{\omega l^3}{24EI}$$

$$y = \frac{\omega x}{24EI}\left(l^3 - 2lx^2 + x^3\right)$$

$$\delta_{max} = \frac{5\omega l^4}{384EI}$$

Figure 4.5: Deflection of beam formula [7]

We now have new bending moment diagram:

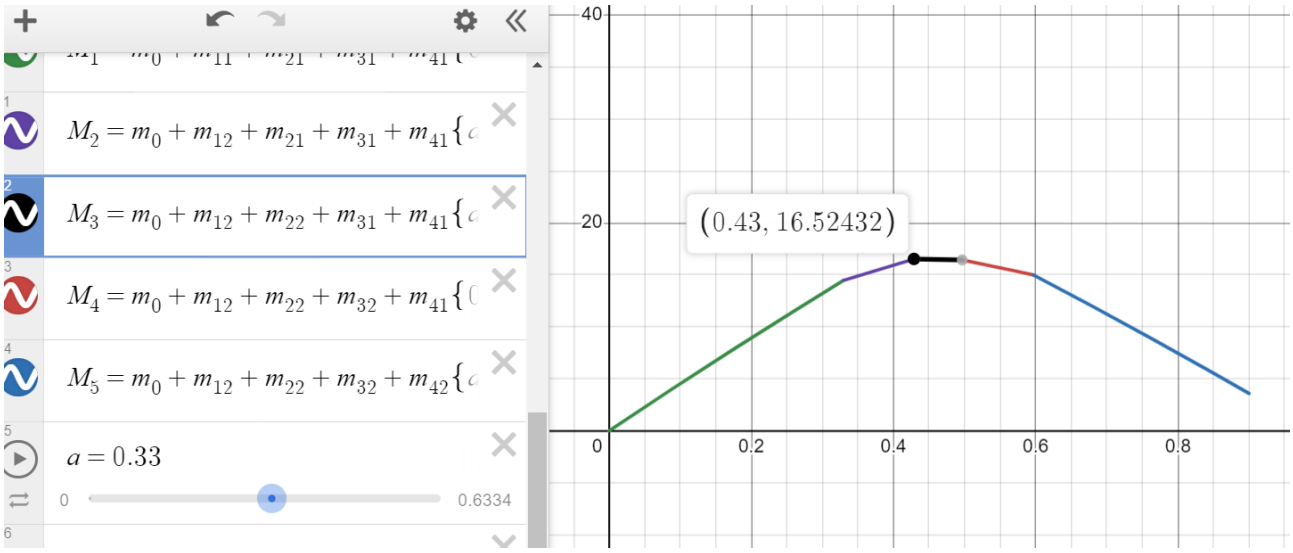


Figure 4.6: Bending moment diagram of simple support case [9]

Now the maximum bending moment is 16.524Nm at $a = 0.33m$

Maximum bending stress is:

$$\sigma_{compression} = -\frac{M_{max}y}{I} = -\frac{16.524 * (72 - 21.938) * 10^{-3}}{3.863 * 10^{-7}} = -2141404.318N < \sigma_{allow}$$

$$\sigma_{tension} = -\frac{M_{max}y}{I} = -\frac{16.524 * (0 - 21.938) * 10^{-3}}{3.863 * 10^{-7}} = 938398.9438N < \sigma_{allow}$$

New shear force diagram:

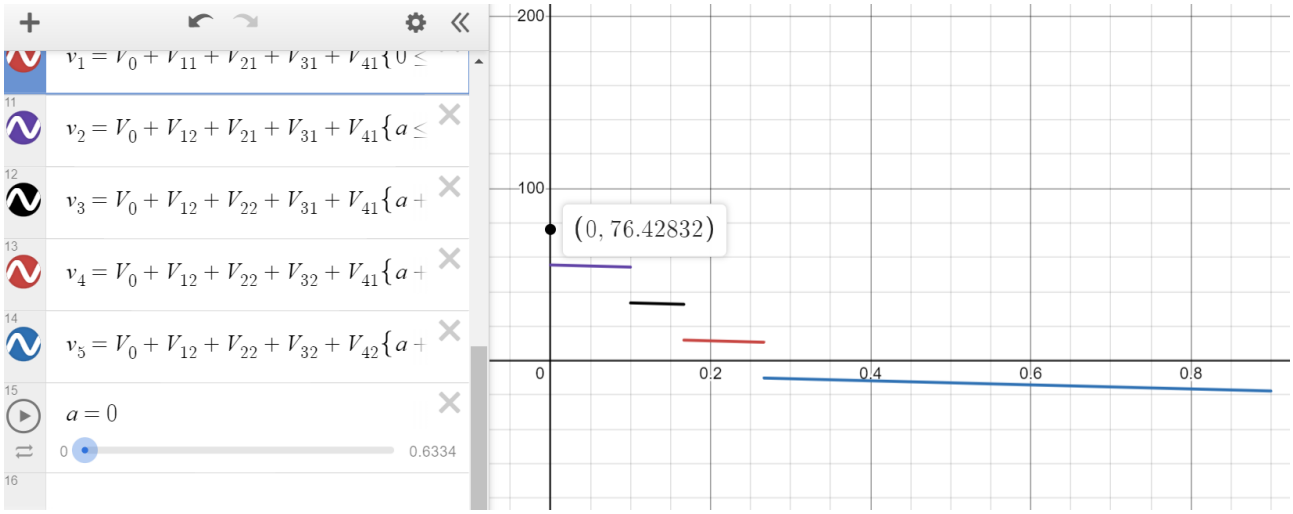


Figure 4.7: Shear moment diagram of simple support [10]

The maximum shear force is now 76.428N at $a = 0m$ or $a = 0.9m$

Maximum shear stress is:

$$\tau_{max} = \frac{V_{max}}{A} = \frac{76.428}{1.014 * 10^{-3}} = 75372.781\ N < \tau_{allow}$$

And deflection of the bridge:

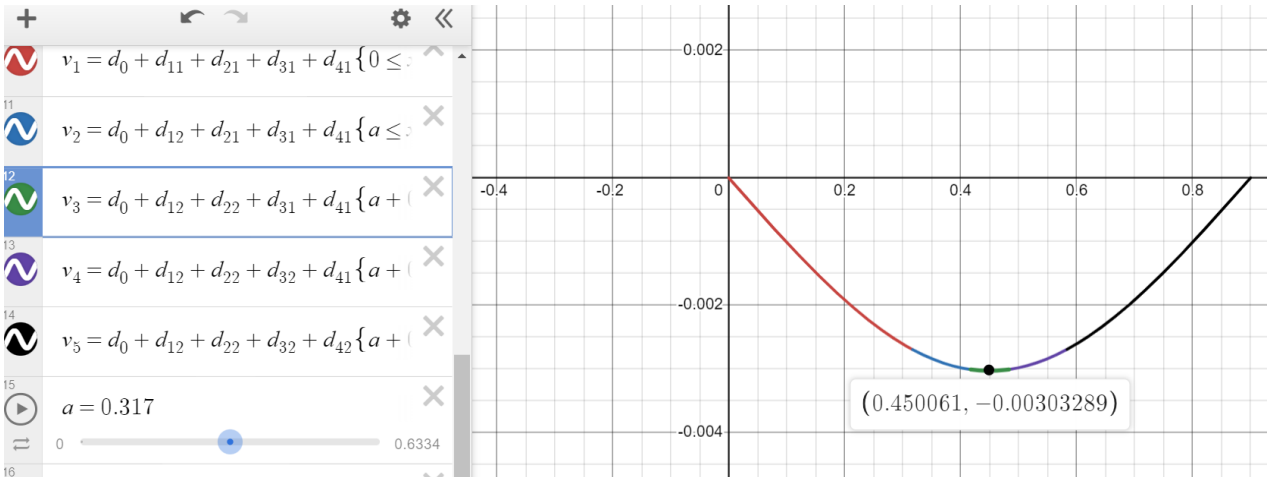


Figure 4.8: Deflection of simple support [11]

Maximum deflection of the bridge is 3.033mm at $a = 0.317m$

If the fixed end of the bridge is replaced with a simple support, the maximum bending stress and shear stress decrease, while the maximum deflection of the bridge increases to approximately three times its original value but still less than the allowable deflection. Nevertheless, the design still satisfies the stiffness condition.

Our bridge had the lowest deflection's value on the demonstration day and satisfied the requirements of project. Without any problem with design, load or bridge damage when the object is moving. Based on what we gain from class, the cross-section, deflection values are different based on the result designed. Especially the deflection result between the calculations and the actual value are not the same.

The reason had the different here because some external things that come into occur during the performance, surface of dynamic track. Calculations represent theoretical based on formula assumptions while actual values are real-world measurements that can vary due to numerous factors. We are not living in a perfect engineering world, everything has its own tolerance, that is why calculations, and actual values often do not match each other. However, the difference between calculations and actual values is not too significant.

4.3. Appendix

Calculate moment of inertia:

Area moment of inertia using parallel axis theorem

$$I = I' + Ad^2$$

Calculate moment of inertia:

Find centroid

Choose Oxy as the sketch

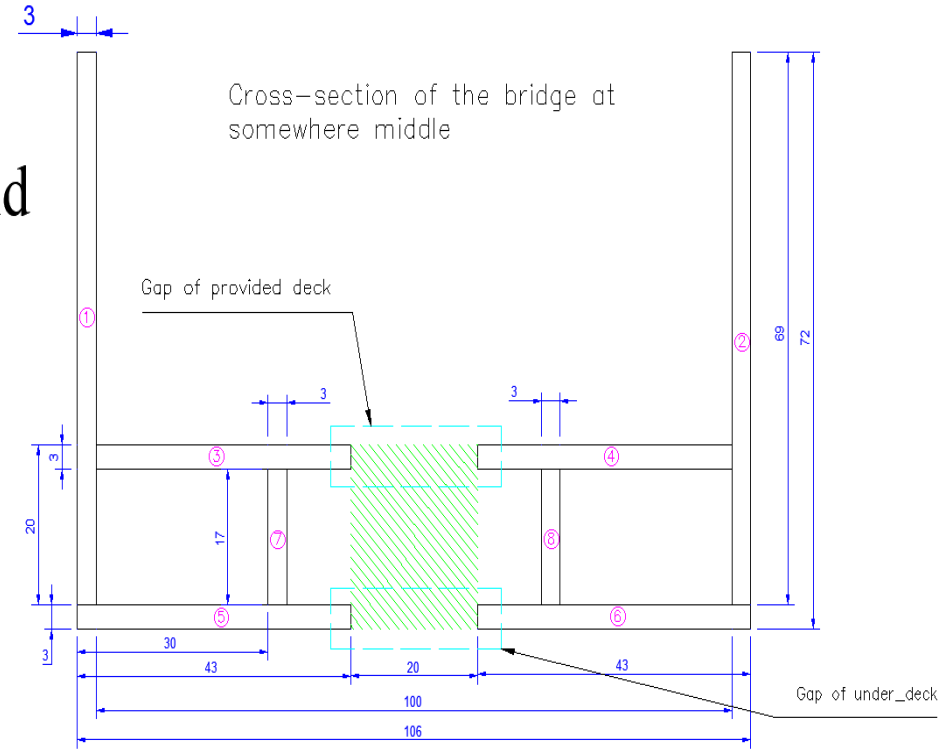


Figure 4.9: Cross-section of the bridge at somewhere middle

$\overline{x_1} = \overline{x_2} = 37.5$

$\overline{x_3} = \overline{x_4} = 21.5$

$\overline{x_5} = \overline{x_6} = 1.5$

$\overline{x_7} = \overline{x_8} = 11.5$

$A_1 = A_2 = 207$

$A_3 = A_4 = 120$

$A_5 = A_6 = 129$

$A_7 = A_8 = 51$Neutral axis is located at:

$$\bar{x} = \frac{\sum_{i=1}^{8} A_i \overline{x_i}}{\sum_{i=1}^{8} A_i} = 21.938mm$$

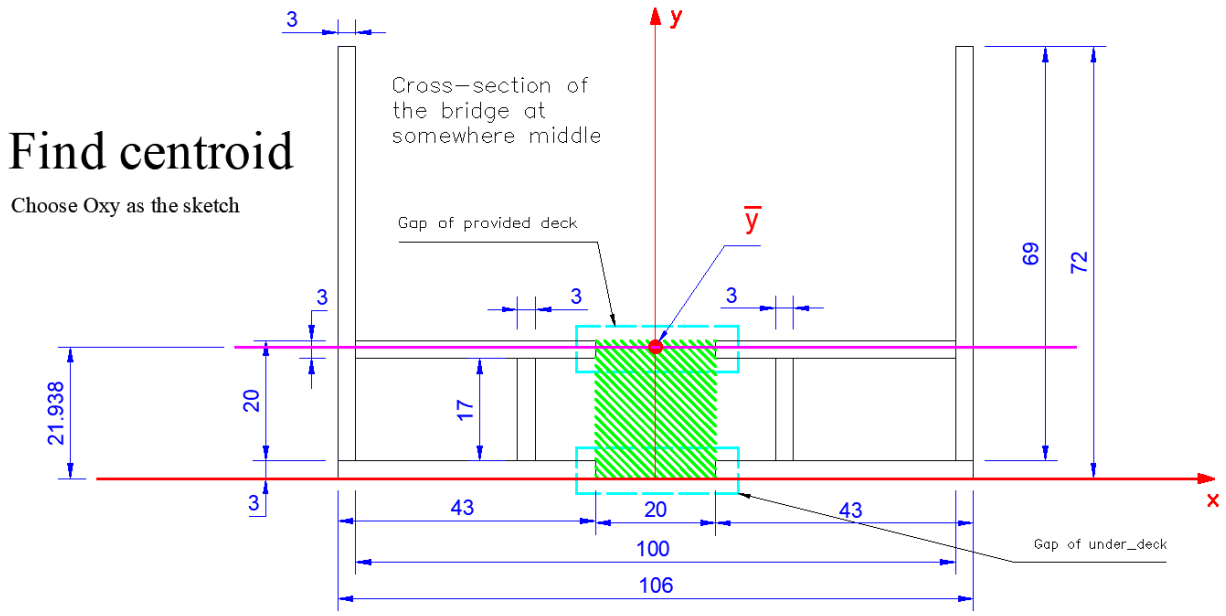


Figure 4.10: The centroid at the cross-section of the bridge

Substituting these variables into area of moment of inertia using parallel axis theorem formula, we have

$$I_1 = I_2 = \frac{1}{12} * 3 * 69^3 + 207 * (37.5 - 21.938)^2 = 132257.6497mm^4$$

$$I_3 = I_4 = \frac{1}{12} * 40 * 3^3 + 120 * (21.5 - 21.938)^2 = 113.0213mm^4$$

$$I_5 = I_6 = \frac{1}{12} * 43 * 3^3 + 129 * (1.5 - 21.938)^2 = 53981.5779mm^4$$

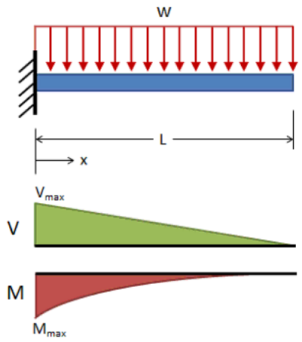
$$I_7 = I_8 = \frac{1}{12} * 3 * 17^3 + 51 * (11.5 - 21.938)^2 = 6784.7940mm^4$$

So, moment of inertia of the cross section is:

$$I = \sum_{i=1}^{8} I_i = 386274.0858mm^4 = 3.863 * 10^{-7}m^4$$

Superposition Method:

Cantilever, Uniform Distributed Load



Deflection:

$$\delta = -\frac{wx^2}{24EI}\left(6L^2 - 4Lx + x^2\right)$$

$$\delta_{max} = \frac{wL^4}{8EI} \qquad @\, x = L$$

Slope:

$$\theta = -\frac{wx}{6EI}\left(3L^2 - 3Lx + x^2\right)$$

$$\theta_{max} = \frac{wL^3}{6EI} \qquad @\, x = L$$

Shear:

$$V = +w\,(L - x)$$

$$V_{max} = +wL \qquad @\, x = 0$$

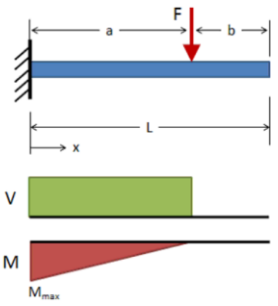
Moment:

$$M = -w\,(L - x)^2 / 2$$

$$M_{max} = -wL^2 / 2 \qquad @\, x = 0$$

Figure 4.11: Distributed Load [3]

Cantilever, Intermediate Load



Deflection:

$$\delta = -\frac{Fx^2}{6EI}(3a - x) \qquad (0 \le x \le a)$$

$$\delta = -\frac{Fa^2}{6EI}(3x - a) \qquad (a \le x \le L)$$

$$\delta_{max} = \frac{Fa^2}{6EI}(3L - a) \qquad @\, x = L$$

Slope:

$$\theta = -\frac{Fx}{2EI}(2a - x) \qquad (0 \le x \le a)$$

$$\theta = -\frac{Fa^2}{2EI} \qquad (a \le x \le L)$$

Shear:

$$V = +F \qquad (0 \le x \le a)$$

$$V = 0 \qquad (a \le x \le L)$$

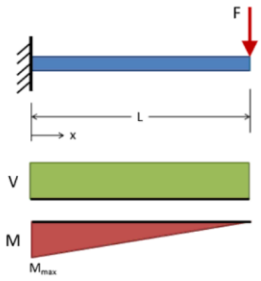
Moment:

$$M = -F\,(a - x) \qquad (0 \le x \le a)$$

$$M = 0 \qquad (a \le x \le L)$$

Figure 4.12: Intermediate Load [4]

Cantilever, End Load



Deflection:

$$\delta = -\frac{Fx^2}{6EI}(3L - x)$$

$$\delta_{max} = \frac{FL^3}{3EI} \qquad @\, x = L$$

Slope:

$$\theta = -\frac{Fx}{2EI}(2L - x)$$

$$\theta_{max} = \frac{FL^2}{2EI} \qquad @\, x = L$$

Shear:

$$V = +F$$

Moment:

$$M = -F\,(L - x)$$

$$M_{max} = -FL \qquad @\, x = 0$$

Figure 4.13: End point load [5]

Density of acrylic: 1.18g/cm^3.

According to Figure 2.2, Figure 2.3, and Figure 2.4. The thickness of each component is 3mm = 0.3cm.

Surface area of each component:

- Rail Bridge: 641cm^2.
- Under_deck: 872.85cm^2.
- Support: 170cm^2.
- Given_deck: 825.43cm^2.

The volume of each component, calculated by [thickness 0.3cm * surface area]:

- Rail Bridge: 192.3cm^3.
- Under_deck: 261.675cm^3.
- Support: 51cm^3.
- Given_deck: 247.629cm^3.

Total mass of the bridge before putting the load on without the weight of two ends of the bridge which are clamped and supported (minus 50mm both side):

$$1.097kg$$

- **Distributed load from weight of the bridge**

$$w = \frac{1.097 * 9.81}{0.9} = 11.9573\ N/m$$

  - Deflection:

$$v_w = \frac{-11.9573x^2}{24 * 1.1 * 10^9 * 3.863 * 10^{-7}}(x^2 - 4 * 0.9 * x + 6 * 0.9^2)\ for\ (0 \le x \le 0.9)$$

  - Shear force and bending moment:

$$V_w = 11.9573(0.9 - x)$$

$$M_w = -11.9573\frac{(0.9 - x)^2}{2}$$

- **Total weight of PAScars and the load:** 0.5kg + 8kg =8.5kg
- **Intermediate load from the 4 wheels $P1 = P2 = P3 = P4 = \frac{8.5 \times 9.81}{4} = 20.85N$**
  - **For P1:**
    - Deflection:

$$v_{P1,1} = \frac{-20.85}{6 * 1.1 * 10^9 * 3.863 * 10^{-7}}a^2(3x - a)\ for\ (a \le x)$$

$$v_{P1,2} = \frac{-20.85}{6 * 1.1 * 10^9 * 3.863 * 10^{-7}}x^2(3a - x)\ for\ (x \le a)$$

    - Shear force and bending moment:

$$V_1 = 20.85 \; for \; (x \leq a)$$

$$M_1 = -20.85(a - x) \; for \; (x \leq a)$$

- **For P2:**
  - Deflection:

$$v_{P2,1} = \frac{-20.85}{6 * 1.1 * 10^9 * 3.863 * 10^{-7}}(a + 0.1)^2(3x - a - 0.1) \; for \; (a + 0.1 \leq x)$$

$$v_{P2,2} = \frac{-20.85}{6 * 1.1 * 10^9 * 3.863 * 10^{-7}}x^2[3(a + 0.1) - x] \; for \; (x \leq a + 0.1)$$

  - Shear force and bending moment:

$$V_{P1,1} = 20.85 \; for \; (x \leq a \; + 0.1)$$

$$M_2 = -20.85(a - x) \; for \; (x \leq a + 0.1)$$

- **For P3:**

$$v_{P3,1} = \frac{-20.85}{6*1.1*10^9*3.863*10^{-7}}(a + 0.1666)^2(3x - a - 0.1666) \; for \; (a + 0.1666 \leq x)$$

$$v_{P3,2} = \frac{-20.85}{6 * 1.1 * 10^9 * 3.863 * 10^{-7}}x^2[3(a + 0.1666) - x] \; for \; (x \leq a + 0.1666)$$

$$V_3 = 20.85 \; for \; (x \leq a + 0.1666)$$

$$M_3 = -20.85(a + 0.1666 - x) \; for \; (x \leq a + 0.1666)$$

- **For P4:**
  - Deflection:

$$v_{P4,1} = \frac{-20.85}{6*1.1*10^9*3.863*10^{-7}}(a + 0.2666)^2(3x - a - 0.2666) \; for \; (a + 0.2666 \leq x)$$

$$v_{P4,2} = \frac{-20.85}{6 * 1.1 * 10^9 * 3.863 * 10^{-7}}x^2[3(a + 0.2666) - x] \; for \; (x \leq a + 0.2666)$$

  - Bending moment and shear force:

$$V_4 = 20.85 \; for \; (x \leq a + 0.2666)$$

$$M_4 = -20.85(a + 0.2666 - x) \; for \; (x \leq a + 0.2666)$$

- **End Point load "By" act on the beam at support B:**

$$v_{By} = \frac{By}{6 \times 1.1 \times 10^9 \times 3.863 \times 10^{-7}}x^2(3 * 0.9 - x)$$

$$V_{By} = -By$$

$$M_{By} = By(L - x)$$

- Using formula from Figure 4.12 we can simplify:
  - $V_1 = 0 \qquad for \; (x \geq a)$
  - $V_2 = 0 \qquad for \; (x \geq a + 0.1)$

- $V_3 = 0$ $for\ (x \geq a + 0.1666)$
- $V_4 = 0$ $for\ (x \geq a + 0.2666)$

and

- $M_1 = 0$ $for\ (x \geq a)$
- $M_2 = 0$ $for\ (x \geq a + 0.1)$
- $M_3 = 0$ $for\ (x \geq a + 0.1666)$
- $M_4 = 0$ $for\ (x \geq a + 0.2666)$

While there is support at B so the deflection at B is 0, which means:

$$v_{By} + v_{P1,1} + v_{P2,1} + v_{P3,1} + v_{P4,1} + v_{P5,1} = 0\ at\ x = 0.9m$$

By substituting v_{By}, $v_{P1,1}$, $v_{P2,1}$, $v_{P3,1}$, $v_{P4,1}$ and $v_{P5,1}$ with inertia $I = 3.863 * 10^{-7}m^4$ into the equations above, we have $By = -57.202a^3 + 131.569a^2 + 36.506a + 7.887$

Therefore:

$$v_{By} = \frac{-57.202a^3 + 131.569a^2 + 36.506a + 7.887}{6 * 1.1 * 10^9 * 3.863 * 10^{-7}} x^2(3 * 0.9 - x)$$

$$V_{By} = -(-57.202a^3 + 131.569a^2 + 36.506a + 7.887)$$

$$M_{By} = (-57.202a^3 + 131.569a^2 + 36.506a + 7.887\ )(0.9 - x)$$

References

References: Including books, technical journals, patents, lectures. (if applicable). Please follow the referencing guide by our RMIT Learning Lab (link). You may also use the Easy Cite referencing tool by our library (link). Improperly cited sources will receive zero credit.

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